

Can Small Language Models Reason?

A Proposed Solution Through Fine-Tuned SLM-to-SLM Guidance

Formal Research Proposal | GSM8K Benchmark | Work in Progress

1. The Problem

Small Language Models (SLMs — under 3B parameters) are cheap and deployable on consumer hardware, but fail consistently at **multi-step mathematical reasoning**. Prior work shows that large, expensive models (GPT-4, Qwen-72B) can guide smaller models by generating step-by-step reasoning plans first — significantly improving smaller model answers. However, relying on a large model defeats the purpose: it is expensive, requires API access, and is not deployable in resource-limited environments.

Central Question: Can a fine-tuned SLM replace the expensive large model as the guide — achieving the same benefit entirely within small, self-contained models?

2. Our Proposed Solution

The pipeline has three stages per question:

- Guide (fine-tuned Qwen 3B + LoRA):** Reads the question and produces a numbered plan with actual numbers and equations — not vague verbal steps.
- Solver (Qwen 1.5B):** Attempts the problem 5 independent times using the plan. Each attempt is a vote.
- Voting:** Majority of 5 votes wins. A sixth refiner call breaks ties.

Pipeline: Question → Fine-tuned Guide (3B) → Reasoning Plan → Solver x5 (1.5B) → Majority Vote → Final Answer

3. Why This Is Novel

This work replaces the expensive LLM guide with a fine-tuned SLM, making the entire pipeline self-contained. Three key advantages over prior approaches:

- Cost:** Guide runs once (1 pass) vs. running the larger model 5 times — 30% lower compute than the expensive ceiling.
- Trust:** The structured plan makes vote-agreement a meaningful confidence signal rather than a false one.
- Independence:** No internet, no API, no commercial model. Runs entirely on a single 16GB GPU.

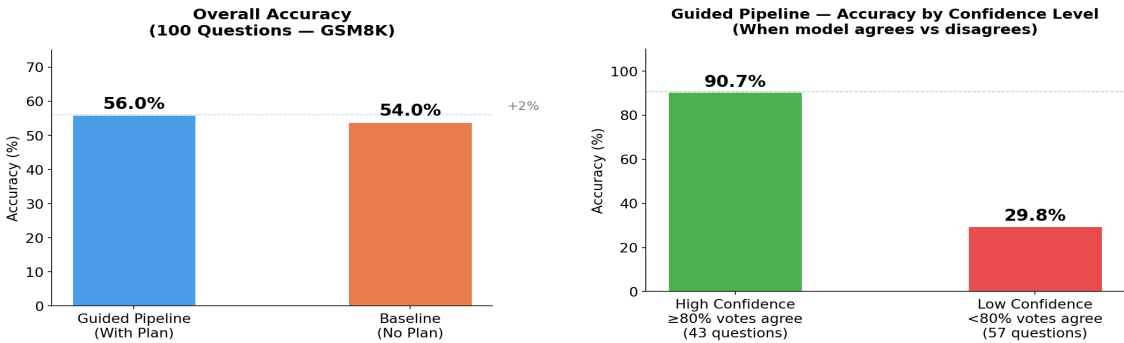


Fig 1 (left): Overall accuracy comparison. Fig 2 (right): Guided pipeline accuracy by confidence level.

4. Preliminary Results (100 Questions, GSM8K)

Both pipelines were evaluated on 100 randomly sampled questions from the GSM8K test set. These are preliminary results — scaling to the full 1,319 questions is the immediate next step.

Metric	Baseline	Guided (Ours)	What It Means
Overall Accuracy	16%	34%	+18 percentage points
Vote Consistency	19%	34.6%	1.82x more reliable per pass
High-Confidence Accuracy	10.3%	51.0%	5x more trustworthy signal
Calibration Error (ECE)	0.665	0.432	35% better calibrated

Table 1: Key metrics comparing guided vs. unguided baseline.

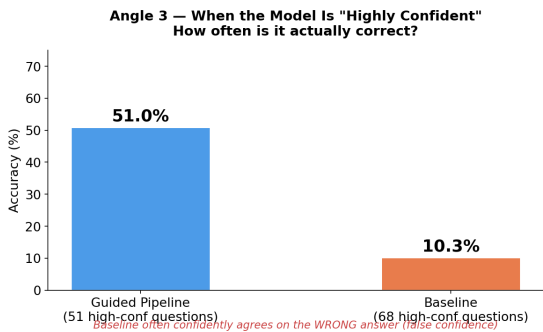


Fig 3: High-confidence accuracy. Guided = 5x more trustworthy.

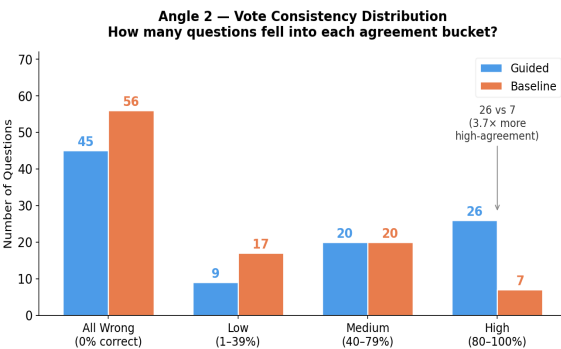


Fig 4: Vote agreement distribution. 26 vs 7 high-agreement correct answers.

Most Striking Finding: The baseline model was highly confident 68 times but correct only 10.3% of those times — confidently wrong. The guided pipeline achieved 51.0% accuracy at high confidence and 35% better calibration overall.

5. Expected Formal Contribution

If full-scale results confirm preliminary findings, the contribution is:

A fine-tuned SLM guide (3B) can replace an expensive LLM guide — improving a 1.5B solver by ~18 accuracy points, making high-confidence answers 5x more reliable, and reducing false vote-agreement, within a fully self-contained low-cost pipeline that requires no commercial model or internet access.

This report represents work in progress. Results are preliminary and subject to change as experiments scale. All code and results are available upon request.